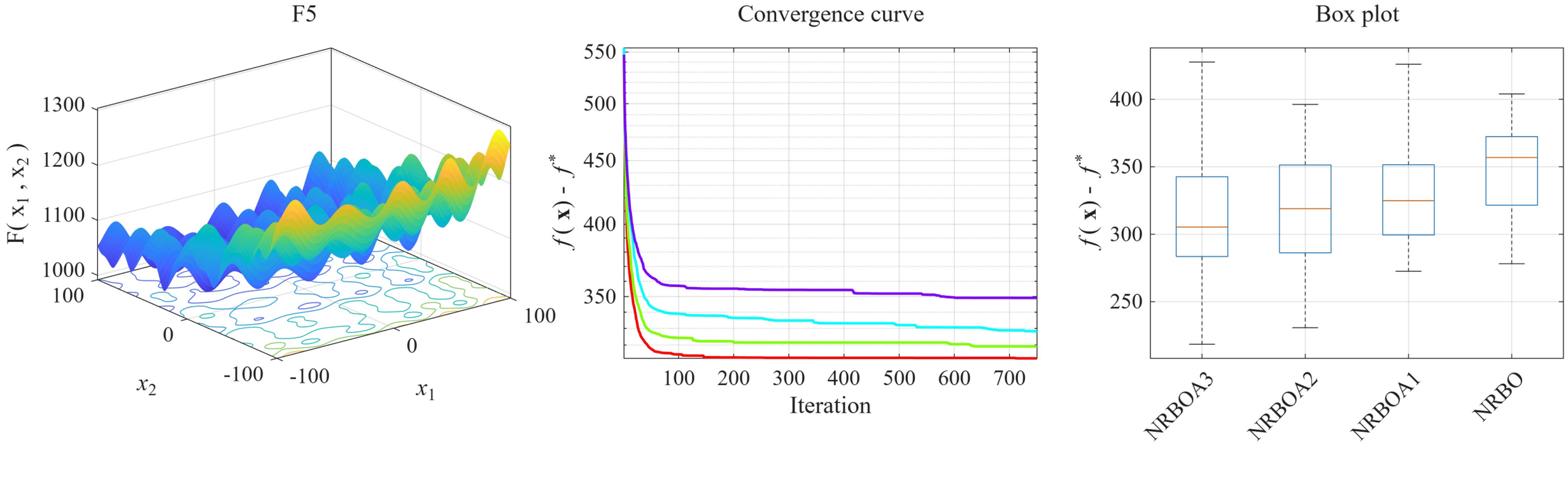

F5
1300
1200
1100
1000
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
550
500
450
400
350
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
400
350
300
250
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO